Inferential statistics

Using sample data to draw conclusions about a population

Inferential statistics this week and next

1. Estimation
2. Significance tests
3. Linear models
4. Multiple testing

Estimating population parameters

Weddell seal dive durations



- Marine biologists want to estimate the mean dive duration of weddell seals
- Estimated population size in Antarctica ~200,000 seals

- They cannot measure every dive of every seal in Antarctica
- They therefore measured dive duration in a sample of 10 seals from the Ross sea by attaching satellite telemetric tags

Weddell seal dive durations



Dive duration (mins)
49.5
46.3
50.2
54.6
45.8
45.8
54.9
50.8
44.7

Dive duration (mins)
49.7

Weddell seal dive durations

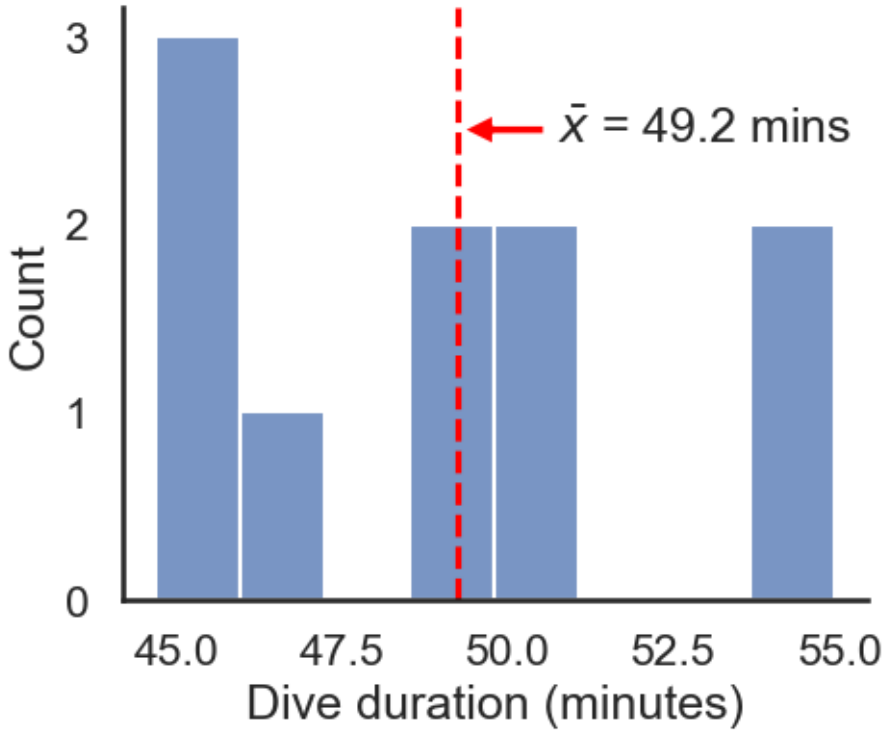


The sample mean is denoted $\bar{x}$

Dive duration (mins)
49.5
46.3
50.2
54.6
45.8
45.8
54.9
50.8

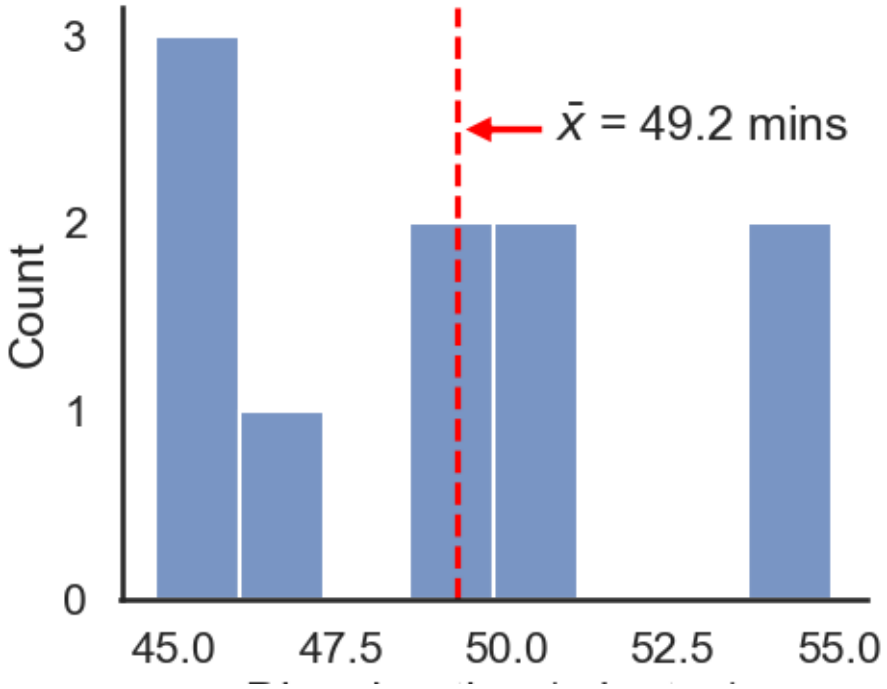
Dive duration (mins)
44.7
49.7
$\bar{x} = \mathbf{49.2}$

Distribution of dive durations



- The mean dive duration of this sample is 49.2 minutes
- But we want to know the overall mean duration of all Weddell seal dives
- This is called the **population mean**, usually denoted μ

Distribution of dive durations



- A **point estimate** of the population mean is usually denoted $\hat{\mu}$
- The hat symbol above $\hat{\mu}$ shows that this is an **estimate** of the population mean μ
- The point estimate of the population mean equals the sample mean:

$$\hat{\mu} = \bar{x} = 49.2 \text{ mins}$$

A point estimate is not enough information

But what if the marine biologists sampled a different set of 10 weddell seal dives

Sample 1
49.5
46.3
50.2
54.6
45.8
45.8
54.9

Sample 1
50.8
44.7
49.7
$\bar{x}_1 = \mathbf{49.2}$

Sample 2
44.7
49.3
48.2
37.4
38.4
44.2
41.9
48.6
42.5
54.3
$\bar{x}_2 = \mathbf{45.0}$

Sample 3
45.7
43.4
47.3
39.9
44.3
44.3
46.5
43.2
45.0
44.0
$\bar{x}_3 = \mathbf{44.4}$

Three different point estimates

A point estimate is not enough information

- We have just one point estimate of the population mean ($\hat{\mu} = 49.2$ minutes)
- But there is some uncertainty in its value because only 10 dives were recorded
- How much uncertainty is there? For example, is it:
  - 49.2 ± 10 mins (around 39 and 59 mins)?
  - 49.2 ± 1 mins (around 48 and 50 mins)?
  - 49.2 ± 0.1 mins (around 49.1 and 49.3 mins)?
- We need to quantify the uncertainty in dive durations

A point estimate is not enough information

- This uncertainty is called **sampling error**
  - So you can ask, how much error is there in the point estimate?
- The inverse of error is **precision**.
  - So you can also ask, how precise is the point estimate?
- In conclusion, the full information we require is:
  - a point estimate of the population mean
  - the error (or precision) associated with that estimate
- What determines the size of the error, and how do we calculate it?

The error in an estimate of a population mean

$$\text{error} = \frac{\text{Standard deviation of the sample}}{\text{Square root of number of observations}}$$

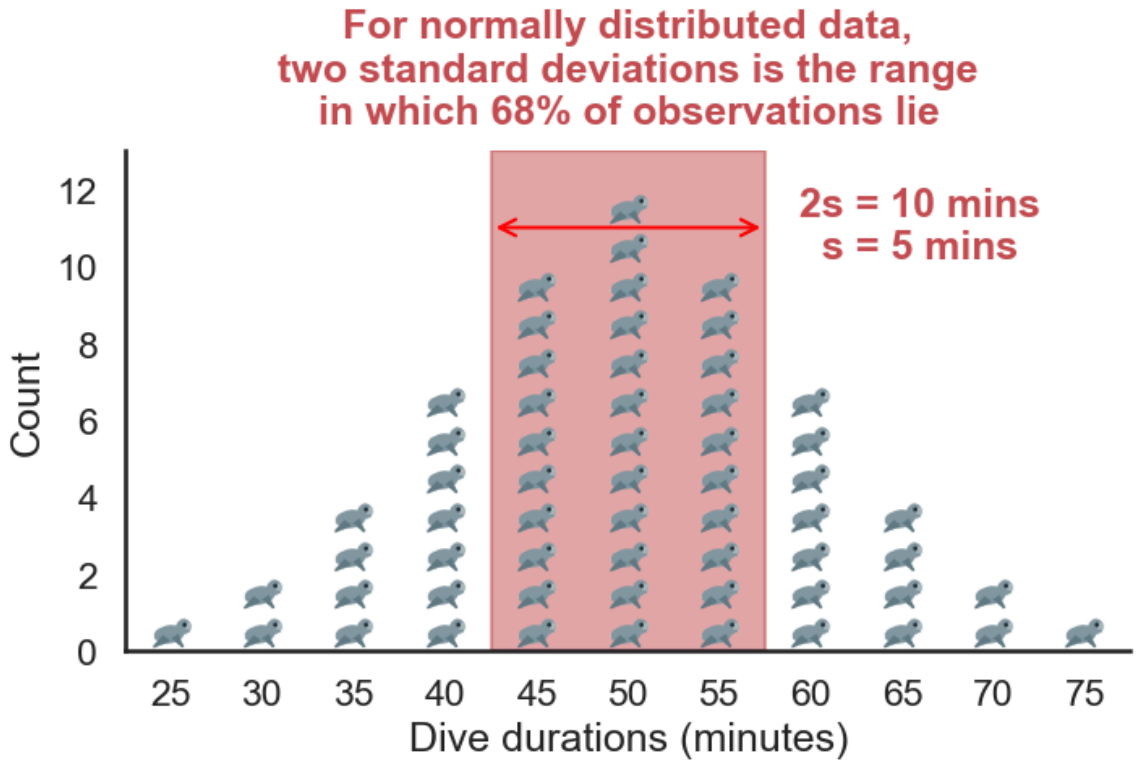
$$\text{error} = \frac{s}{\sqrt{n}}$$

Where does this formula come from?

Reminder: How do we measure variation?

- Variation is how spread out the data are around the mean
- There are three common measures of variation:
  - **Range**: maximum value minus minimum value
  - **Inter quartile range**: the range containing the middle 50% of data
  - **Standard deviation** (usually denoted s)

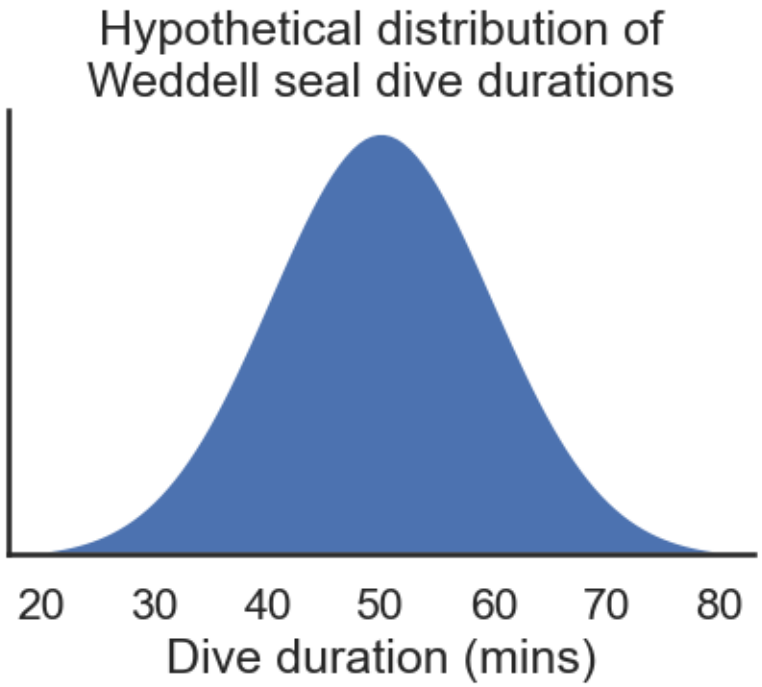
Reminder: What is one standard deviation?



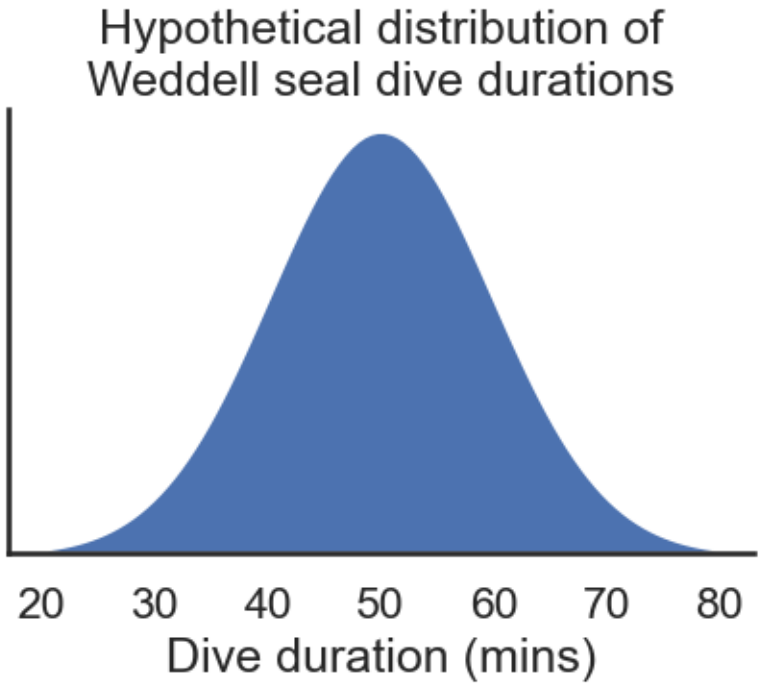
A thought experiment

Let's assume

- dive durations are normally distributed (bell-shaped)
- the population mean dive duration of weddell seals (μ) is 50 minutes
- the population variation (standard deviation) in dive duration (σ) is 10 minutes



Monte-Carlo simulation



- Sample 100 random dive durations from this normal distribution

```
from numpy.random import normal

x_1 = normal(loc=50, scale=10, size=100)

print(x_1.round(1))
```

```
[61.2 59.9 44.2 30.9 49.6 39.6 50.8 30.  61.1 44.1 54.  50.7 49.4 52.  33.9 72.1
29.3 71.5 41.7 50.5
 60.2 44.6 43.1 53.3 52.3 55.2 56.6 51.6 29.6 43.6 52.  55.4 48.6 52.3 56.1 37.1
47.7 46.8 47.6 69.6
 55.7 43.8 66.3 44.7 54.6 60.3 56.3 67.7 43.8 64.1 39.5 60.8 53.1 47.7 54.5 36.6
61.  50.5 53.6 31.9
 58.6 46.6 43.5 45.8 42.8 51.6 35.8 33.8 71.7 38.3 58.9 33.  33.9 56.1 55.6 70.3
62.3 60.7 55.  48.6
 39.  48.  67.1 78.4 54.7 53.8 27.8 62.2 39.1 36.8 60.1 41.3 45.8 50.1 60.8 39.7
61.4 40.6 47.6 45.2]
```

Monte-Carlo simulation


```
[61.2 59.9 44.2 30.9 49.6 39.6 50.8 30.  61.1 44.1 54.  50.7 49.4 52.  33.9 72.1
29.3 71.5 41.7 50.5
 60.2 44.6 43.1 53.3 52.3 55.2 56.6 51.6 29.6 43.6 52.  55.4 48.6 52.3 56.1 37.1
47.7 46.8 47.6 69.6
 55.7 43.8 66.3 44.7 54.6 60.3 56.3 67.7 43.8 64.1 39.5 60.8 53.1 47.7 54.5 36.6
61.  50.5 53.6 31.9
 58.6 46.6 43.5 45.8 42.8 51.6 35.8 33.8 71.7 38.3 58.9 33.  33.9 56.1 55.6 70.3
62.3 60.7 55.  48.6
 39.  48.  67.1 78.4 54.7 53.8 27.8 62.2 39.1 36.8 60.1 41.3 45.8 50.1 60.8 39.7
61.4 40.6 47.6 45.2]
```


Calculate the mean of the 100 dive durations

```
xbar_1 = x_1.mean()

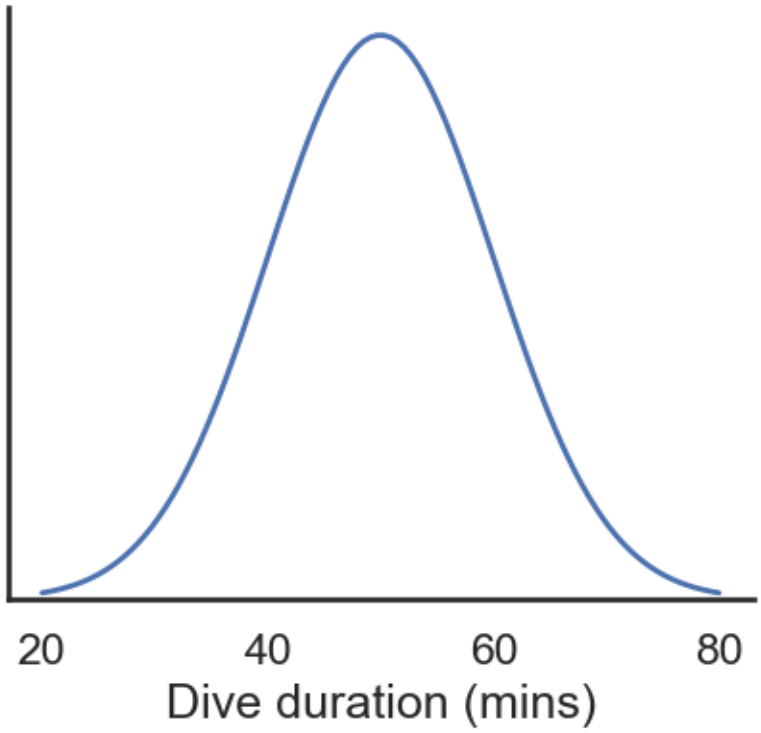
print(f'xbar = {xbar_1:.1f} mins')
```

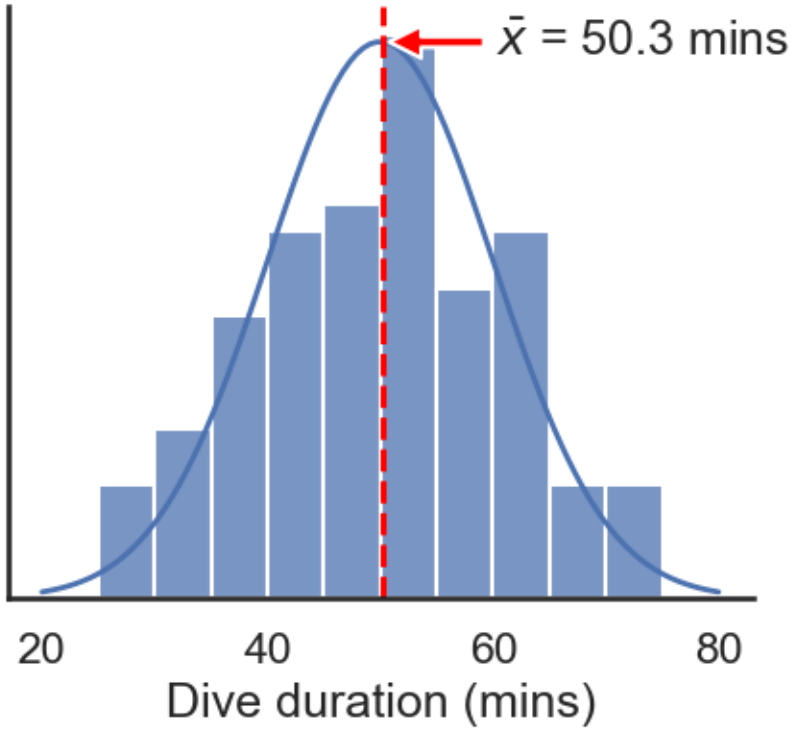
```
xbar = 50.3 mins
```

Monte-Carlo simulation

Compare hypothetical population distribution with the random sample of 100 dives

- Population mean $\mu = 50$ mins, population st. dev. $\sigma = 10$ mins
- Point estimate of population mean is $\hat{\mu} = \bar{x} = 49.9$ mins





Let's repeat this 200 times

Create 200 samples of n = 100 random dive durations each from a normal distribution with mean 50 and standard deviation 10

```
x_200 = normal(loc=50, scale=10, size=(200, 100))
```

Calculate the 200 means of the 200 samples

```
xbar_200 = x_200.mean(axis=1)

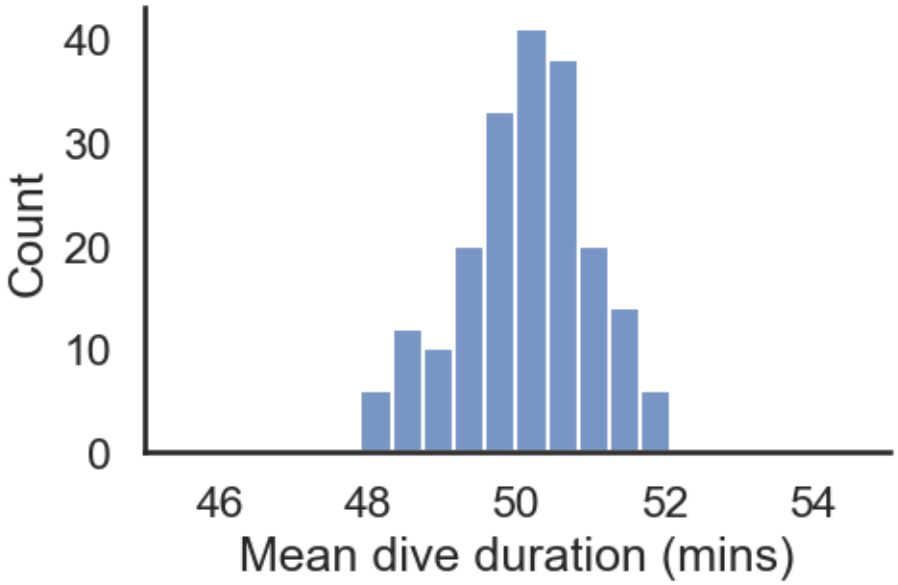
print(xbar_200.round(1))
```

```
[49.7 49.9 48.6 50.  49.8 50.1 50.  49.2 48.5 49.  50.3 49.3 49.3 49.7 48.2 50.1
49.4 50.7 49.6 49.2
 48.4 50.6 50.3 51.1 49.3 49.8 49.6 48.8 50.5 49.3 49.5 50.6 50.7 48.6 50.2 51.
51.4 48.7 50.  49.6
 50.6 49.7 51.4 50.3 50.8 51.  50.5 50.4 49.8 50.5 49.5 49.2 51.1 50.8 49.9 50.
50.7 50.5 51.3 51.2
 49.2 48.7 50.  50.1 50.4 49.  51.3 50.5 50.5 50.3 49.2 51.9 48.3 49.  50.4 50.7
50.8 48.5 49.8 52.
 50.1 51.2 49.1 51.4 49.  48.9 50.2 50.7 50.2 49.6 49.1 51.2 50.9 50.4 51.1 49.6
```

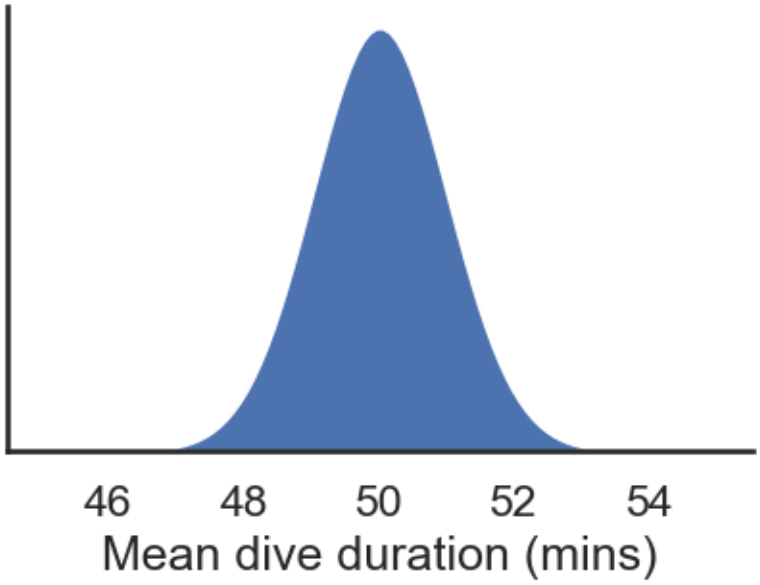
```
51.2 50.2 50.3 49.8
 49.7 49.6 50.1 50.4 50.3 50.5 50.9 51.2 51.  51.6 50.5 48.6 48.6 49.3 48.1 49.7
50.8 49.1 50.2 50.4
 48.3 49.6 50.4 49.9 51.1 50.7 51.4 49.8 50.  52.  50.2 48.5 50.7 50.4 50.  50.8
50.1 49.4 50.7 51.5
 50.2 49.9 49.8 49.2 51.6 50.3 49.8 50.  51.3 49.2 49.5 50.2 51.3 50.1 51.5 50.2
50.5 51.5 49.6 50.2
 50.3 47.9 50.8 50.5 49.6 50.6 49.5 50.1 50.4 51.8 48.1 48.6 51.3 50.6 50.2 50.
50.2 48.2 50.4 52.1
 50.2 50.9 49.7 50.8 49.4 50.3 50.8 49.7 50.9 48.9 50.7 50.7 51.7 49.6 51.2 50.6
50.  49.7 49.3 49.6]
```

A histogram of the 200 sample means

```
[49.7 49.9 48.6 50.  49.8 50.1 50.  49.2 48.5 49.  50.3 49.3 49.3 49.7 48.2 50.1
49.4 50.7 49.6 49.2
 48.4 50.6 50.3 51.1 49.3 49.8 49.6 48.8 50.5 49.3 49.5 50.6 50.7 48.6 50.2 51.
51.4 48.7 50.  49.6
 50.6 49.7 51.4 50.3 50.8 51.  50.5 50.4 49.8 50.5 49.5 49.2 51.1 50.8 49.9 50.
50.7 50.5 51.3 51.2
 49.2 48.7 50.  50.1 50.4 49.  51.3 50.5 50.5 50.3 49.2 51.9 48.3 49.  50.4 50.7
50.8 48.5 49.8 52.
 50.1 51.2 49.1 51.4 49.  48.9 50.2 50.7 50.2 49.6 49.1 51.2 50.9 50.4 51.1 49.6
51.2 50.2 50.3 49.8
 49.7 49.6 50.1 50.4 50.3 50.5 50.9 51.2 51.  51.6 50.5 48.6 48.6 49.3 48.1 49.7
50.8 49.1 50.2 50.4
 48.3 49.6 50.4 49.9 51.1 50.7 51.4 49.8 50.  52.  50.2 48.5 50.7 50.4 50.  50.8
50.1 49.4 50.7 51.5
 50.2 49.9 49.8 49.2 51.6 50.3 49.8 50.  51.3 49.2 49.5 50.2 51.3 50.1 51.5 50.2
50.5 51.5 49.6 50.2
 50.3 47.9 50.8 50.5 49.6 50.6 49.5 50.1 50.4 51.8 48.1 48.6 51.3 50.6 50.2 50.
50.2 48.2 50.4 52.1
 50.2 50.9 49.7 50.8 49.4 50.3 50.8 49.7 50.9 48.9 50.7 50.7 51.7 49.6 51.2 50.6
50.  49.7 49.3 49.6]
```



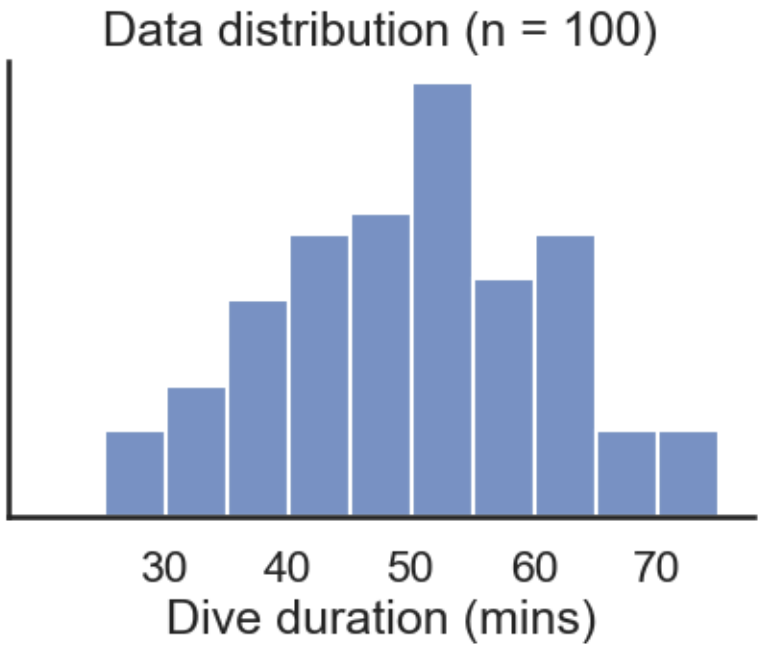
Now sample 100 dives an infinite number of times

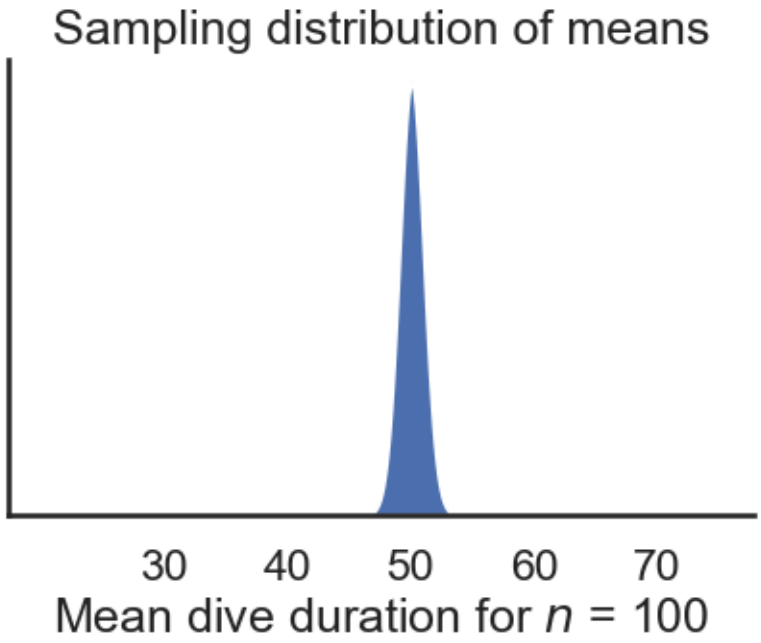


This is called a **sampling distribution**

Do not confuse a data distribution with its sampling distribution

- **Data distribution** is a histogram of the data from an experiment
- **Sampling distribution** is a hypothetical distribution of sample means of sample size *n*
  - we don't know what it is
  - we don't need to know what it is
  - but it is a useful concept for thinking about the error in our estimates

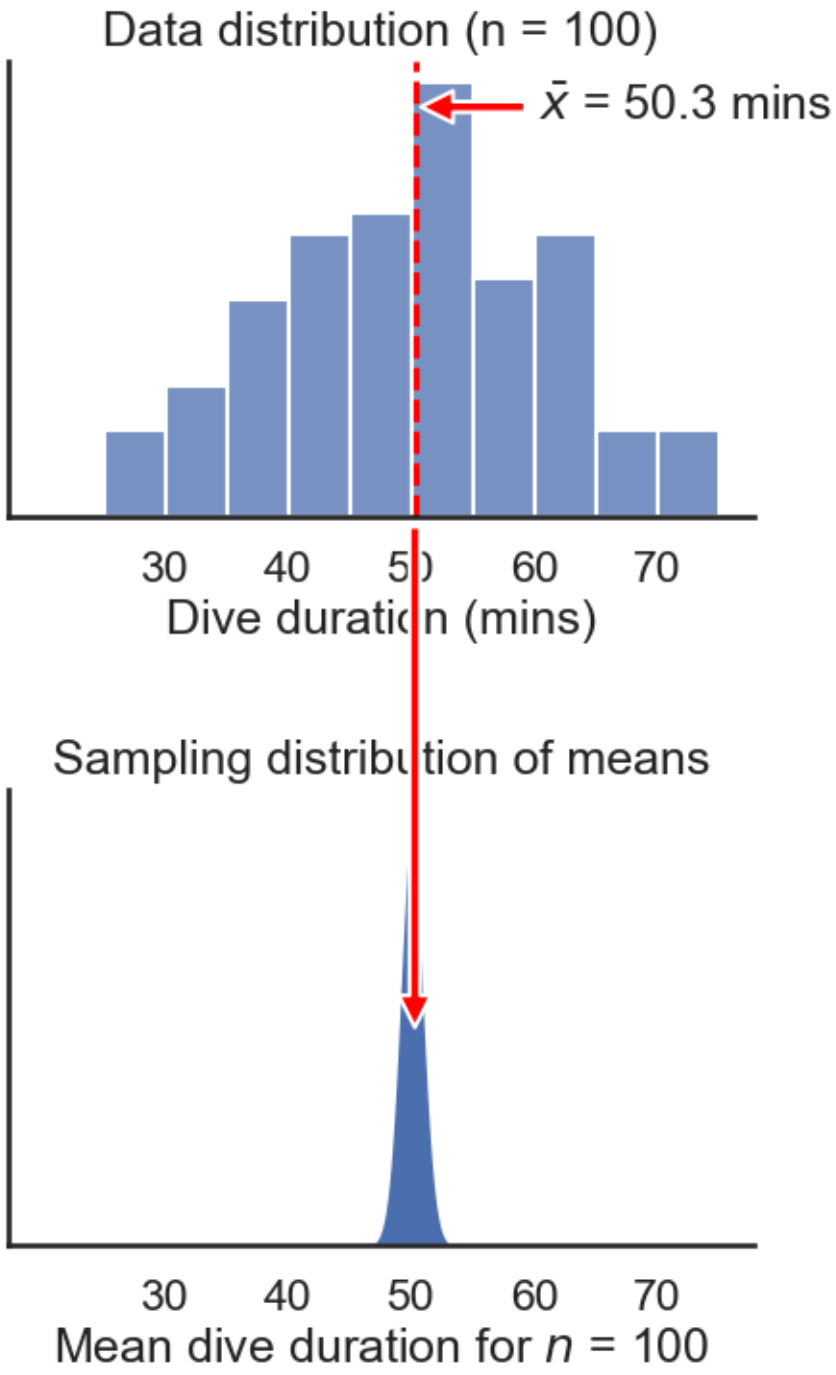




The mean of our single sample lies within the sampling distribution

- The **width** of the sampling distribution therefore tells us how much uncertainty (error) there is in our point estimate
- By convention we use the standard deviation of the sampling distribution as a measure of its width

- Therefore the error in our point estimate is the standard deviation (width) of the sampling distribution



Standard error of the mean

The technical name for the standard deviation of the sampling distribution is the **Standard error of the mean (SEM)**

- "standard" because standard deviation is used to measure variability
- "error" because it tells us the error in our point estimate of the mean, $\hat{\mu}$

Formula for the SEM

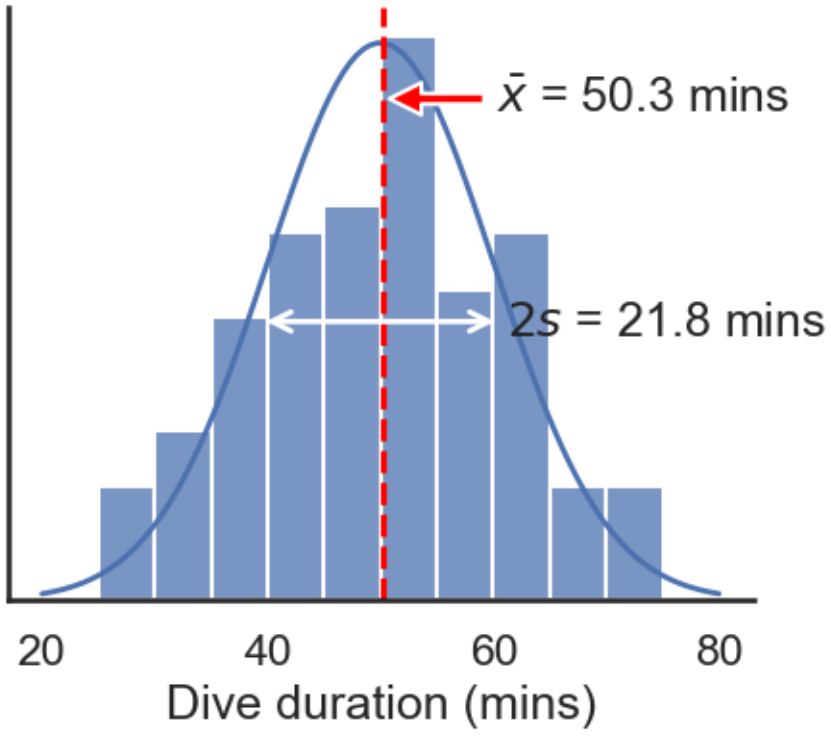
It turns out, and you can prove this mathematically, that

$$\text{SEM} = \frac{\text{Population standard deviation of dive durations}}{\text{Square root of number of observations}} = \frac{\sigma}{\sqrt{n}}$$

But there's a problem

$$\text{SEM} = \frac{\text{Population standard deviation of dive durations}}{\text{Square root of number of observations}} = \frac{\sigma}{\sqrt{n}}$$

- We don't actually know the value of the **population** standard deviation of dive durations, σ
- But we do know the **sample** standard deviation of dive durations, *s*.



Solution

$$\text{SEM} = \frac{\text{Population standard deviation of dive durations}}{\text{Square root of number of observations}} = \frac{\sigma}{\sqrt{n}}$$

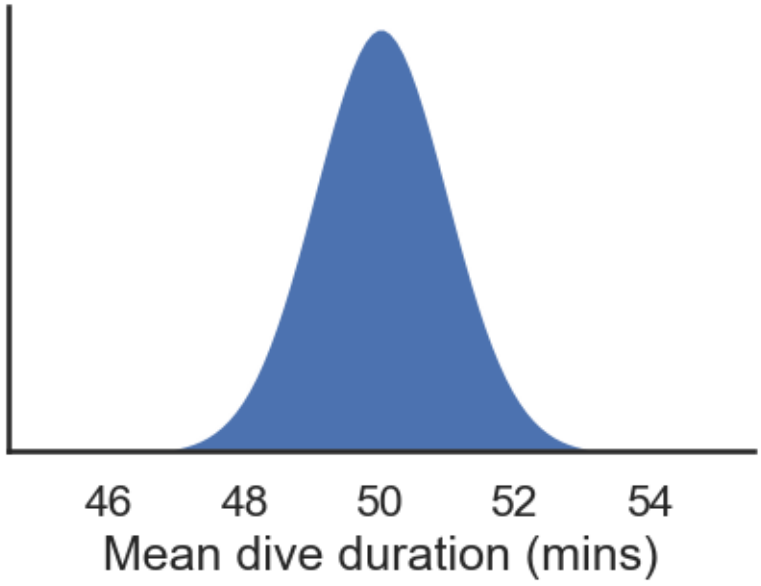
- The sample standard deviation (s) is an estimate of the population standard deviation (σ)
- Use s instead of σ to calculate SEM:

$$\text{SEM} \approx \frac{\text{Sample standard deviation of observed dive durations}}{\text{Square root of number of observations}} = \frac{s}{\sqrt{n}}$$

End of the thought experiment

What have we learnt?

- We pretended to sample our population an infinite number of times
- For each sample we calculated its mean
- We plotted the distribution of these means
- This distribution is called a sampling distribution



End of the thought experiment

What have we learnt?

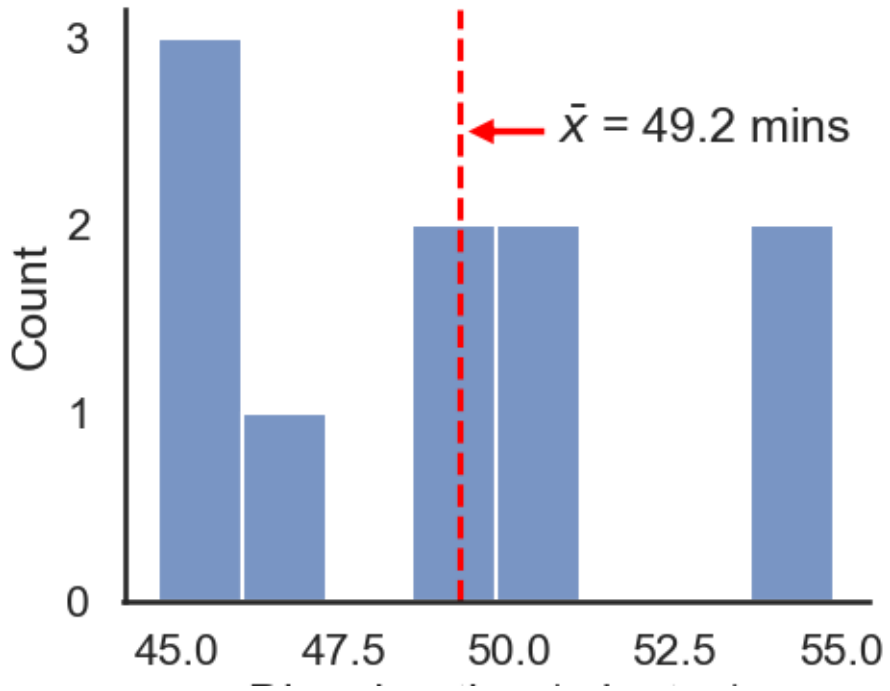
- The width of the sampling distribution tells us how much error there is in the point estimate of the population mean
- We use the standard deviation of the sampling distribution as a measure of its width
- This is called the standard error of the mean
- Because we don't know the population standard deviation we use its estimate, the sample standard deviation
- The formula for the standard error of the mean is

$$\text{SEM} \approx \frac{\text{Sample standard deviation of observed dive durations}}{\text{Square root of number of observations}} = \frac{s}{\sqrt{n}}$$

Interactive plot of the sampling distribution

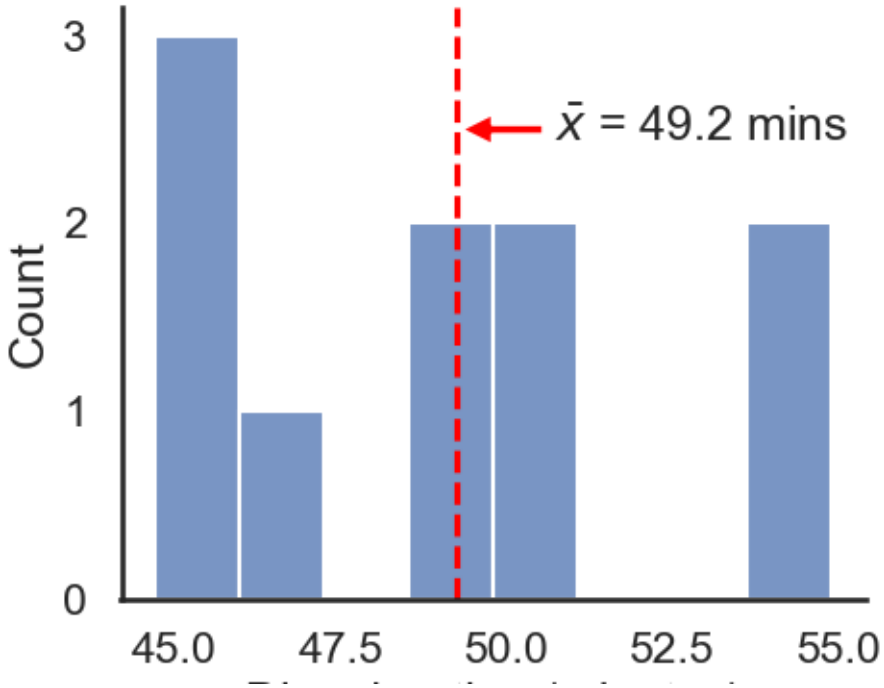
The notebook estimate.ipynb has an interactive plot of the sampling distribution.

Back to the original data of dive durations



Dive duration (mins)
49.5
46.3
50.2
54.6
45.8
45.8
54.9
50.8
44.7
49.7
$\bar{x} =$ **49.2**

The final estimate with error




```
dives = pd.read_csv('../../Datasets/weddell_seal_dives.csv')

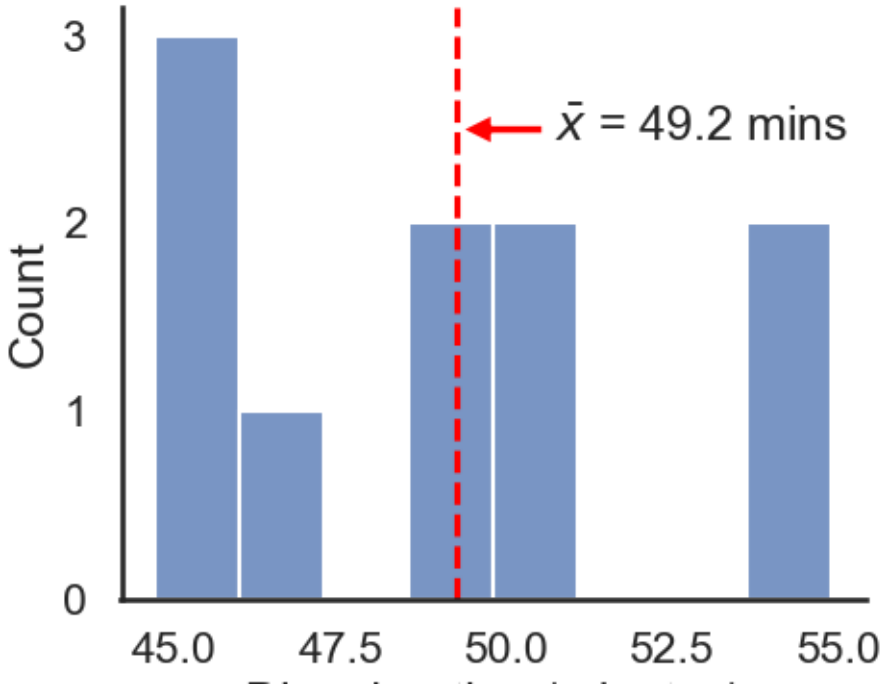
n = dives['dive_duration_min'].count()
xbar = dives['dive_duration_min'].mean()
s = dives['dive_duration_min'].std()

# Either
sem = s/np.sqrt(n)
# or
sem = dives['dive_duration_min'].sem()

print(f'n: {n}')
print(f'mean: {xbar:.2f} mins')
print(f'std: {s:.2f} mins')
print(f'SEM: {sem:.2f} mins')
```

```
n: 10
mean: 49.23 mins
std: 3.61 mins
SEM: 1.14 mins
```

The final estimate with error



The estimated mean dive duration of Weddell seals is

$\hat{\mu} = 49.2 \pm 1.1$ mins

Average dive durations of Weddell seals is 49 minutes plus or minus 1 minute.

A common alternative: the 95% CI

$$\hat{\mu} = \bar{x} \pm \text{SEM}$$

is just one way of reporting a point estimate of the population mean with error.

$$\hat{\mu} = \bar{x} \pm t_{n-1} \times \text{SEM}$$

is another common way and is called a 95% confidence interval (CI)

A common alternative: the 95% CI

$$\hat{\mu} = \bar{x} \pm t_{n-1} \times \text{SEM}$$

The factor t_{n-1} is a correction factor for small sample sizes.

n	t_{n-1}
2	12.71
5	2.78

n	t_{n-1}
10	2.26
20	2.09
50	2.01
100	1.98
500+	1.96

A common alternative: the 95% CI

A rule of thumb:
For samples with more than 20 observations t_{n-1} is about 2, i.e.,

$$\hat{\mu} = \bar{x} \pm 2 \times \text{SEM}$$

For large enough samples, the 95% CI is about two times larger than the standard error.

A common alternative: the 95% CI

The code to calculate t_{n-1} is

```
from scipy.stats import t

t_df = t.ppf(0.975, df=9)

print(f't_df = {t_df:.2f}')
```

```
t_df = 2.26
```

`df` stands for "degrees of freedom". In this case it is one less than the sample size.

A common alternative: the 95% CI

For n = 10, $t_9 = 2.26$

95% CI are usually written as a range from the low bound to the high bound.

- low bound = $\bar{x} - t_{n-1} \times \text{SEM}$
- high bound = $\bar{x} + t_{n-1} \times \text{SEM}$

```
ci_low = xbar - t_df * sem
ci_high = xbar + t_df * sem

print(f'95% CI: [{ci_low:.1f}, {ci_high:.1f}] mins')
```

```
95% CI: [46.6, 51.8] mins
```

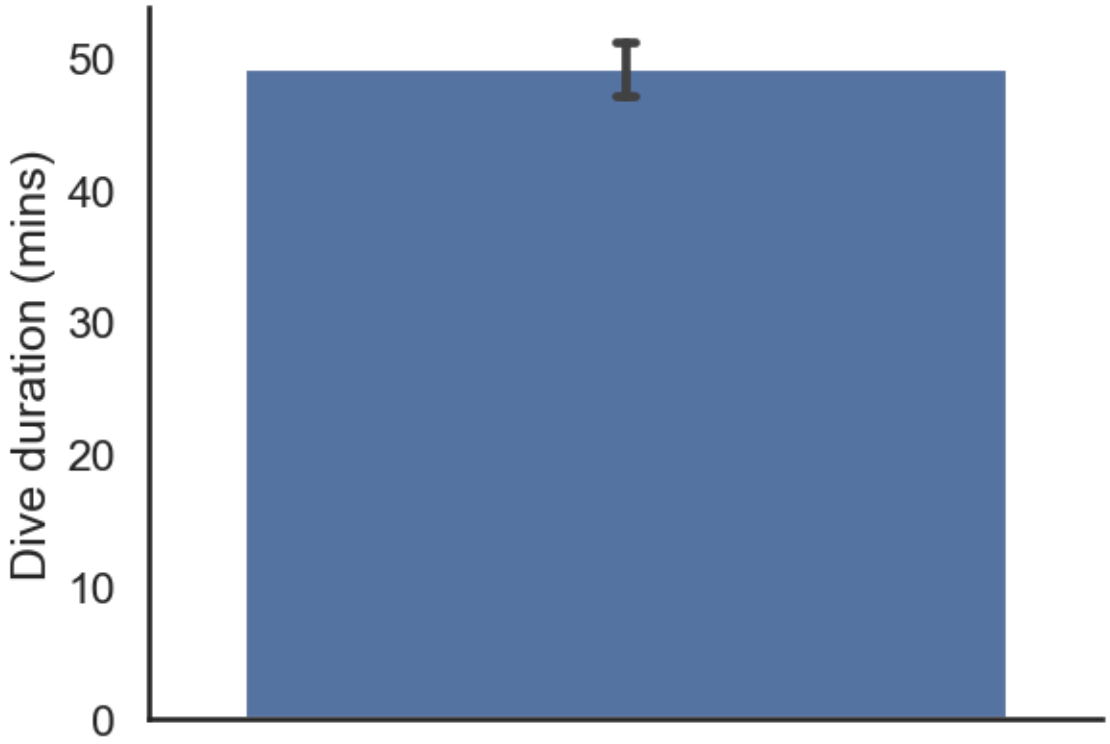
A common alternative: the 95% CI

- The 95% CI of dive durations is [46.6, 51.8] mins.
- Many statistical software packages automatically calculate SEM and 95% CIs for you.

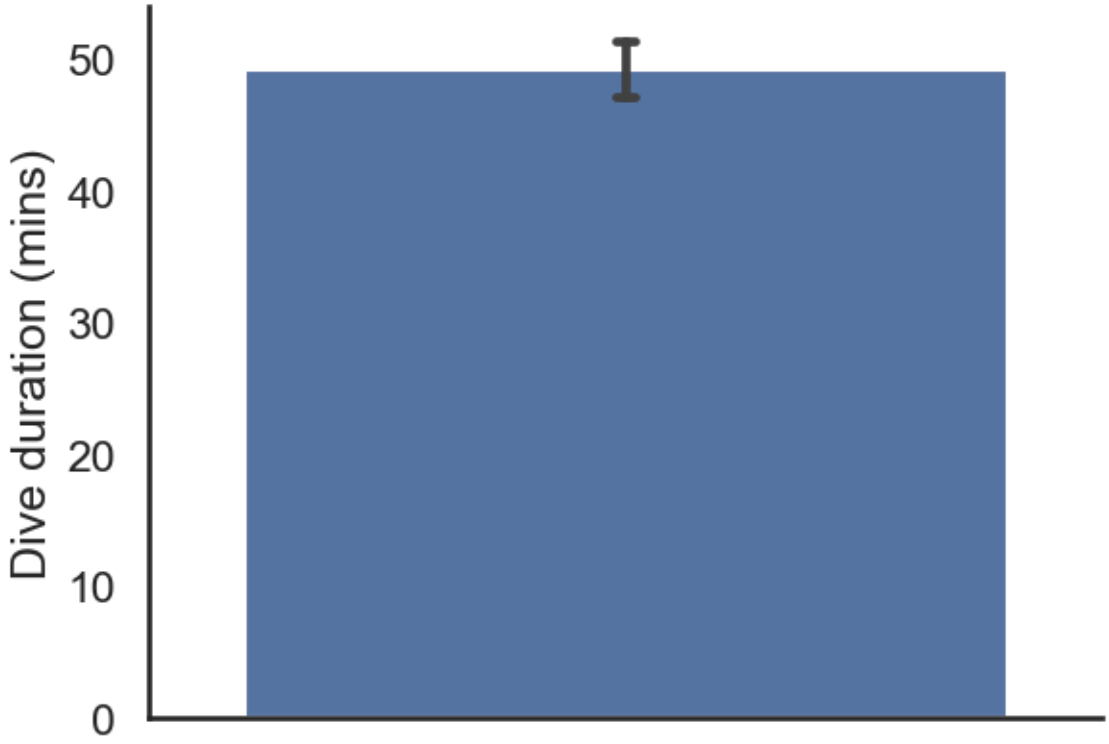
Visualising point estimates and their errors

- Errors are graphically represented as error bars
- Barplots are a common method

```
ax = sns.barplot(data=dives, y='dive_duration_min', capsize=.02);
ax.set_ylabel('Dive duration (mins)');
```



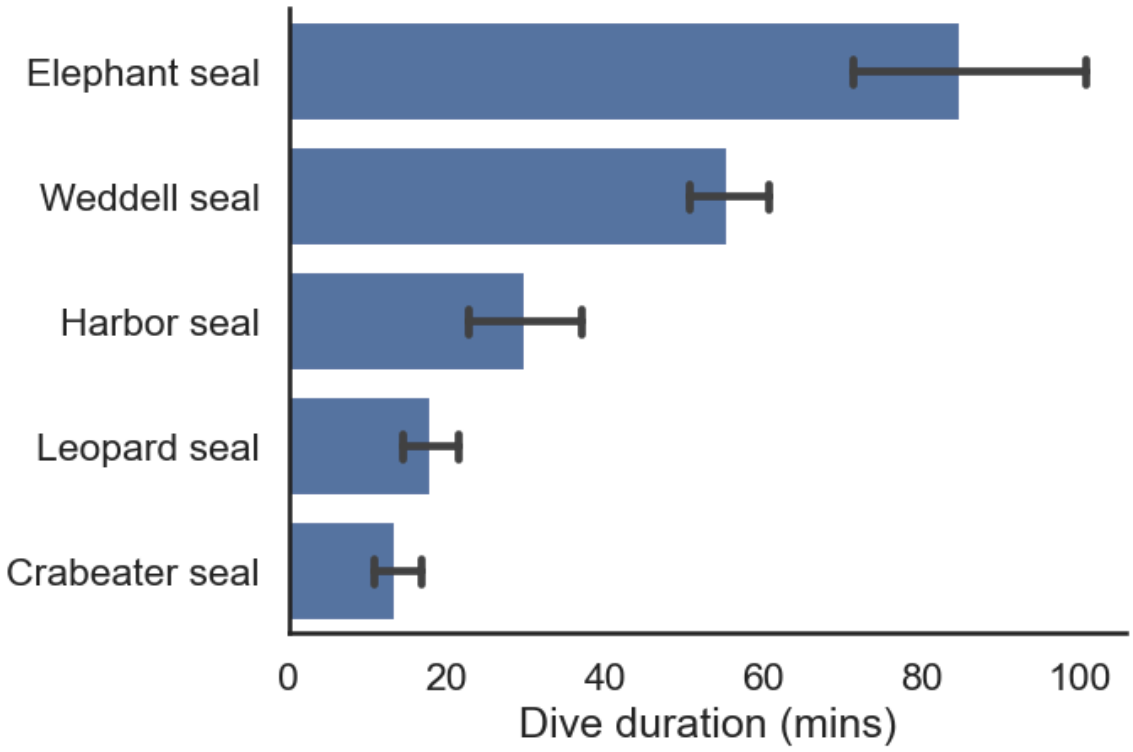
Error bars in a bar plot



- A bar plot creates a bar with the bottom at zero and the top at the point estimate. The errorbar extends from the low bound to the high bound of the 95% CI.
- If there is not too many data points it is also a good idea to include them in the plot so that people can see the position and spread of the raw data.

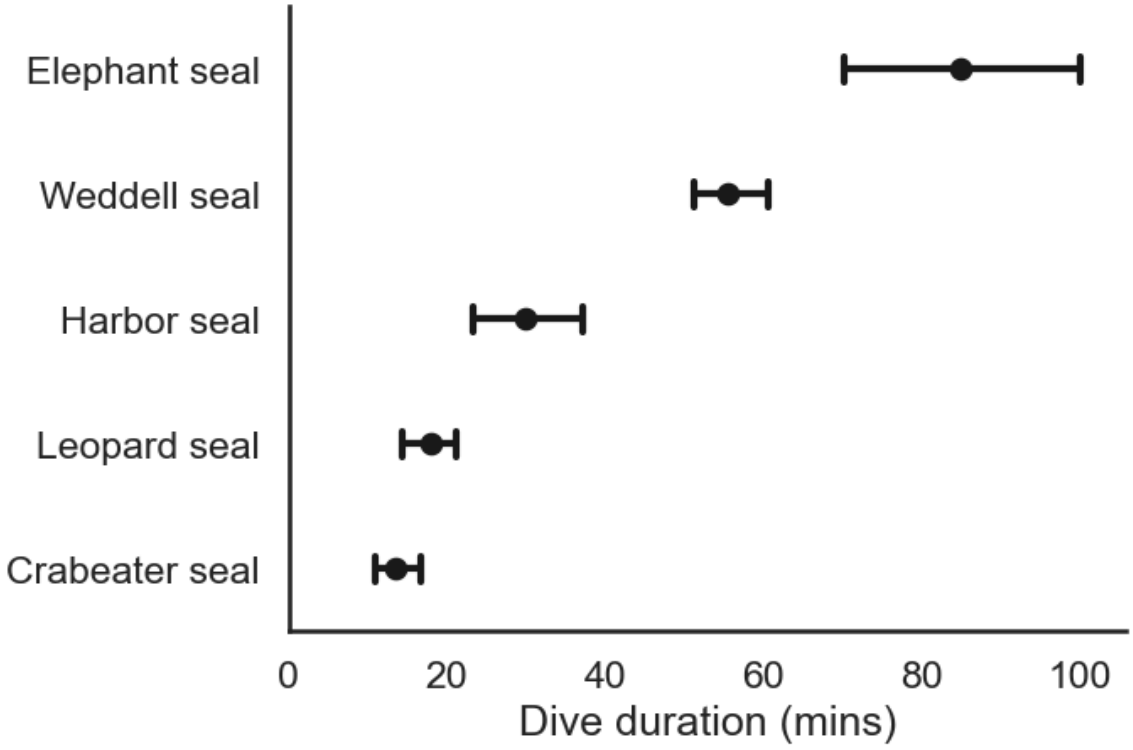
Error bars in a bar plot

These plots are more useful when comparing things. Dive durations of different seal species, for example.



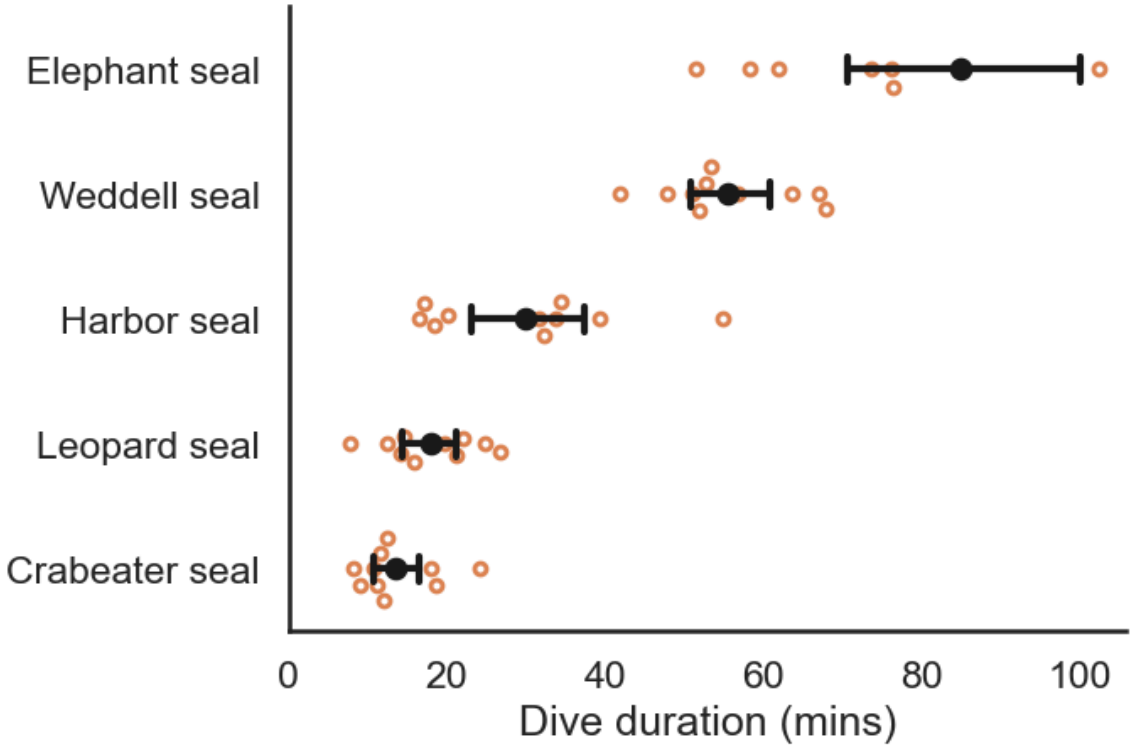
Error bars in a point plot

A point plot just plots a circle at the point estimate. This gives a less cluttered figure than a bar plot.



Include the data

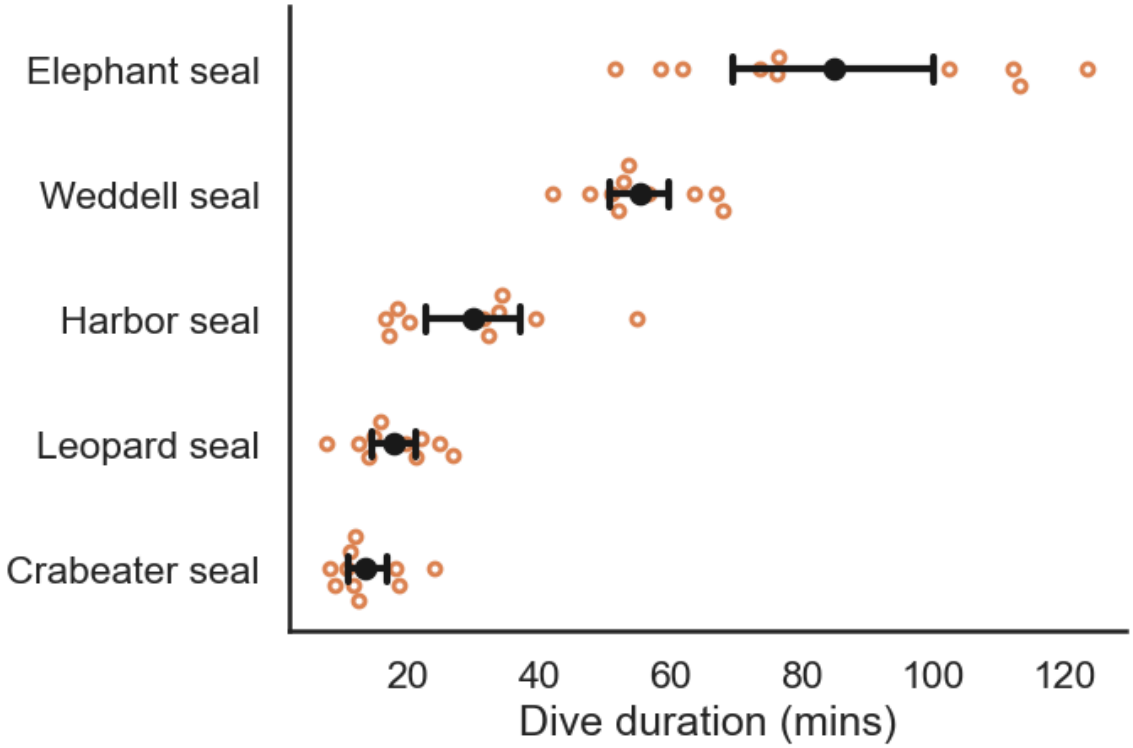
If there are not too many data including them can be useful to see the distribution of the data



Error bars in a point plot

! Important

Always state in the figure caption what errorbars represent (SEM, 95% CI) so that readers know how to interpret them.



This week's class notebook

Three questions:

1. **Weddell seal dive durations**:
   - The same analysis as in the lecture but with a larger data set
2. **Evolution of ground finch beak depths**:
   - Estimate the **difference** in the means of two populations
   - This is a little more complicated and is a prelude to Friday's material
3. **Proportion of hospital patients with vancomycin-resistant intestinal *Enterococcus* bacteria**
   - How to estimate a population proportion from a sample of data
   - Method 1: By using a formula for the standard error of the proportion
   - Method 2: By using Monte-Carlo simulation of the sampling distribution of a proportion

```
2.1.1
```

```
(1.0521172029234378,
 0.9981260180050887,
 1.0521172029234378,
```

```
0.33270867266836285,
0.33270867266836285)
```